문제 E1 — 풀이

구리 막대에서의 열전도 (10점)

0	1	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8	9

파트 A: 짧은 구리 막대 (3.9점)

A.1 (0.2 pt)

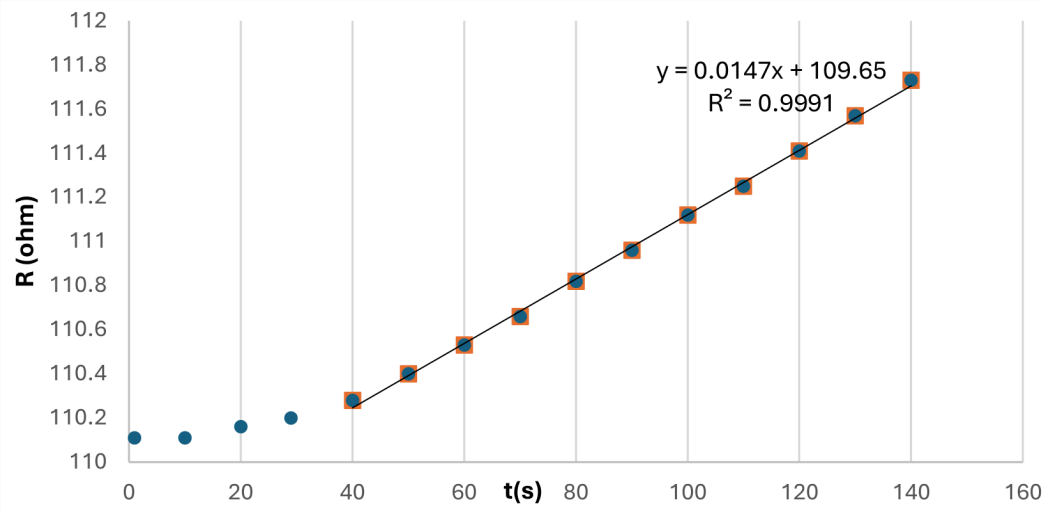
$$R_{en} = 110.11 \pm 0.01\ \Omega \quad \theta_{en} = \frac{R - R_0}{R_0 \alpha}$$

$$\theta_{en} = 25.87 \pm 0.03\ ^\circ\mathrm{C}$$

A.2 (0.5 pt)

n	$R\,(\Omega)$	$t\,(\mathrm{s})$
1	110.11	1
2	110.11	10
3	110.15	20
4	110.18	30
5	110.26	40
6	110.4	50
7	110.53	60
8	110.66	70
9	110.82	80
10	110.96	90
11	111.12	100
12	111.25	110
13	111.41	120
14	111.57	130
15	111.73	140

A.3 (0.8 pt)



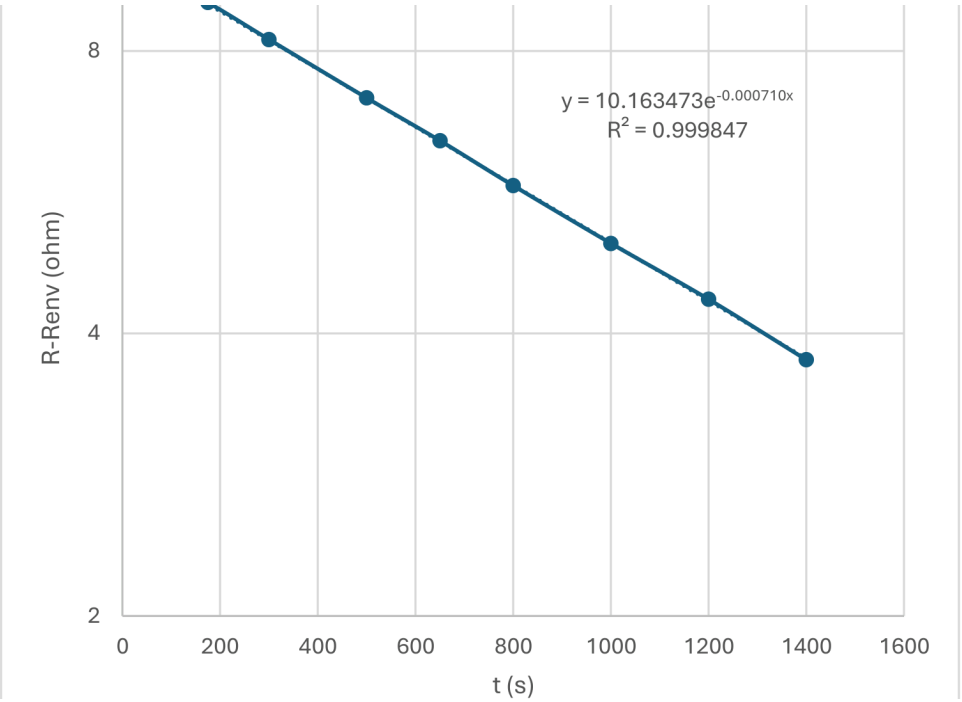
$\Delta R/\Delta t$	$0.0147\ \Omega/\mathrm{s}$	$\dfrac{\Delta\theta}{\Delta t}=\dfrac{1}{R_0\alpha}\dfrac{\Delta R}{\Delta t}$
$\Delta\theta/\Delta t$	$0.03761\ ^\circ\mathrm{C}/\mathrm{s}$	
C_s	$52\pm 2\ \mathrm{J}/^\circ\mathrm{C}$	

$$C_s=\frac{P_1}{\frac{\Delta\theta}{\Delta t}},\quad \Delta C_s\approx C_s\frac{\Delta P_1}{P_1}$$

A.4 (0.5 pt)

n	$R\,(\Omega)$	$t\,(\mathrm{s})$	$(R-R_{en})\,(\Omega)$
1	119.98	40	9.82
2	119.17	175	9.01
3	118.39	300	8.23
4	117.29	500	7.13
5	116.58	650	6.42
6	115.91	800	5.75
7	115.15	1000	4.99
8	114.51	1200	4.35
9	113.91	1400	3.75
10	113.40	1600	3.24

A.5 (0.7 pt)

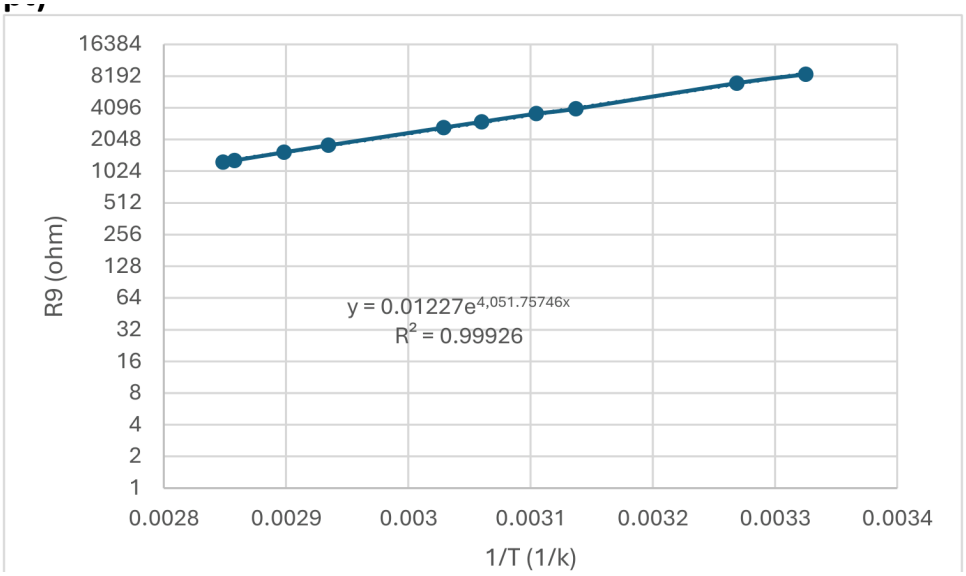


$$\gamma = (710 \pm 3) \times 10^{-6}\ \mathrm{s}^{-1}$$

A.6 (0.5 pt)

n	$R_9\,(\Omega)$	$R\,(\Omega)$	$T\,(\mathrm{K})$	$\frac{1}{T}\,(1/\mathrm{K})$
1	8528	110.78	300.733	0.003325
2	7020	112.81	305.9272	0.003269
3	4005	117.83	318.772	0.003137
4	3601	119.13	322.0984	0.003105
5	3014	120.96	326.7808	0.00306
6	2658	122.27	330.1328	0.003029
7	1803	126.42	340.7515	0.002935
8	1547	128.09	345.0245	0.002898
9	1296	130.00	349.9117	0.002858
10	1246	130.45	351.0631	0.002848

A.7 (0.7 pt)



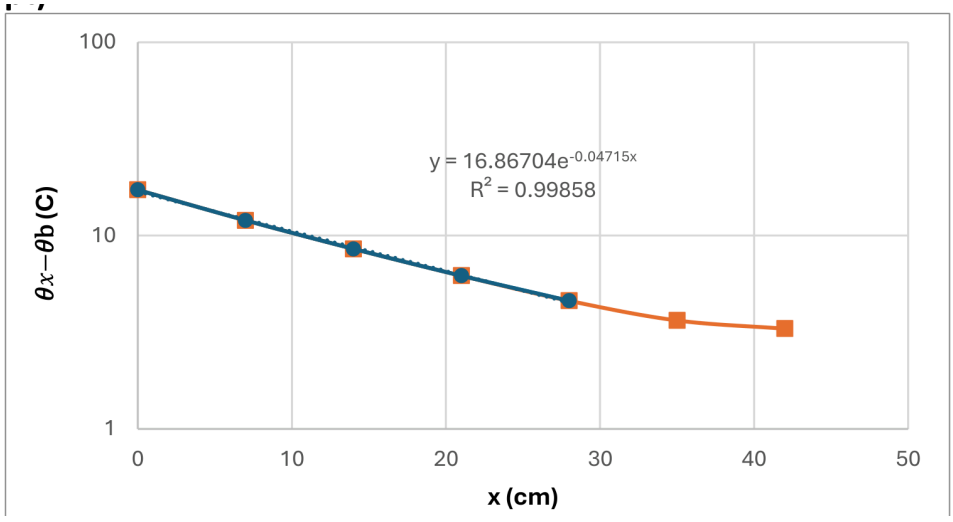
$E_g = 0.698 \pm 0.007\ \text{eV}$

파트 B: 긴 구리 막대 (4.1점)

B.1 및 B.5

$\theta_b = 24.41\ ^\circ\text{C}$

B.2 (0.4 pt)



B.3 (0.6 pt) **B.4 (0.4 pt)**

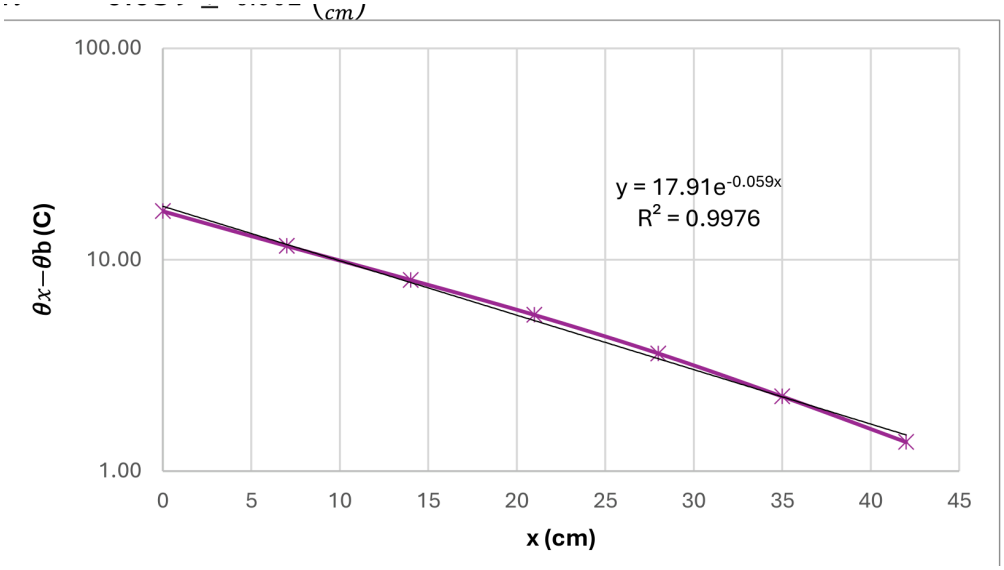
$A^{(0)} = 16.87\ ^\circ\text{C}$ $B = Ae^{-2\lambda d}$

$\lambda^{(0)} = 0.047 \pm 0.005\ (1/\text{cm})$ $B^{(1)} = 0.266\ ^\circ\text{C}$

B.6 (1.0 pt)

$A^{(1)} = 17.91\,^{\circ}\mathrm{C}$

$\lambda^{(1)} = 0.059 \pm 0.002\ (1/\mathrm{cm})$



B.7 (0.9 pt)

$$P_2 = \int_{-0.5\ \mathrm{cm}}^{42.5\ \mathrm{cm}} 2\pi r h(\theta_x - \theta_b)\,dx \quad \text{또는} \quad \int_0^d 2\pi r h(\theta_x - \theta_b)\,dx = \frac{2\pi r h A}{\lambda}\left(1 - e^{-2\lambda d}\right)$$

$$h = \frac{P_2 \lambda}{2\pi r A\left(1 - e^{-2\lambda d}\right)}, \quad k = \frac{2h}{\lambda^2 r}, \quad P_2 = 4.5\ \mathrm{W}$$

$\lambda = 0.053 \pm 0.008\ (1/\mathrm{cm}), \quad h = 0.0037 \pm 0.0003\ \mathrm{W/(K\,cm^2)}, \quad k = 3.9 \pm 0.3\ \mathrm{W/(K\,cm)}$

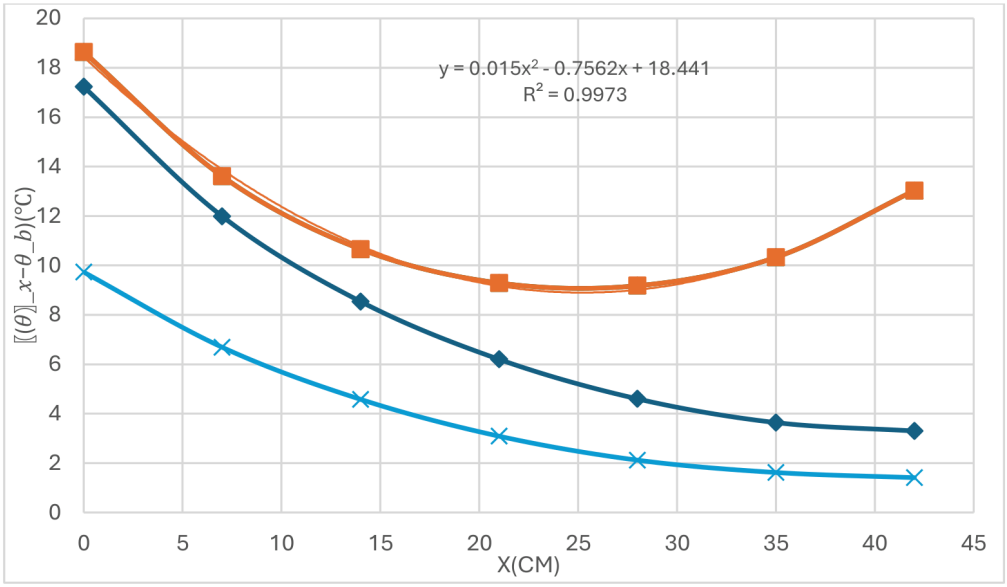
파트 C: 미지의 출력 측정 (2.0점)

C.1 (0.4 pt)

$\theta_b = 27.38\,^{\circ}\mathrm{C}$

n	$x\ (\mathrm{cm})$	$\theta_x\ (^{\circ}\mathrm{C})$	$(\theta_x - \theta_b)\ (^{\circ}\mathrm{C})$	B-1	C-B	역순
1	0	46.02	18.64	17.23	1.41	9.73
2	7	40.99	13.61	11.99	1.62	6.69
3	14	38.04	10.66	8.54	2.12	4.58
4	21	36.67	9.29	6.2	3.09	3.09
5	28	36.56	9.18	4.6	4.58	2.12
6	35	37.71	10.33	3.64	6.69	1.62
7	42	40.41	13.03	3.3	9.73	1.41

C.2 (0.6 pt)



(파란 점들은 그릴 필요가 없다)

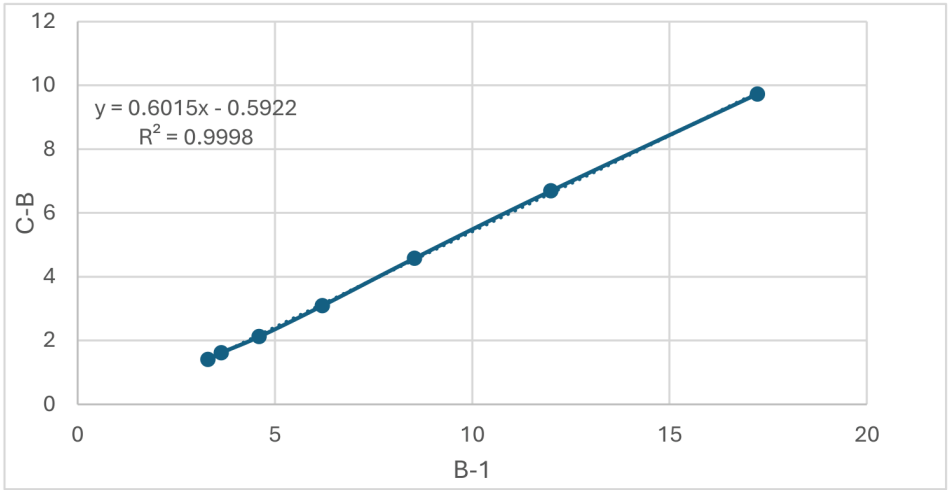
C.3 (1.0 pt)

1) $$\frac{P_3}{P_2} = \frac{\left.\frac{d\theta}{dx}\right|_{x=42\ \mathrm{cm}}}{\left.\frac{d\theta}{dx}\right|_{x=0\ \mathrm{cm}}} = \frac{\left.\frac{\Delta\theta}{\Delta x}\right|_{x=42\ \mathrm{cm}}}{\left.\frac{\Delta\theta}{\Delta x}\right|_{x=0\ \mathrm{cm}}} = \frac{\theta_7 - \theta_6}{\theta_0 - \theta_1}$$

2) $$\frac{P_3}{P_2} = \frac{\left.\frac{d\theta}{dx}\right|_{x=42\ \mathrm{cm}}}{\left.\frac{d\theta}{dx}\right|_{x=0\ \mathrm{cm}}} = \frac{\sinh(\lambda(42 - x_0))}{\sinh(\lambda x_0)}$$

3) 적분법 또는 여러 구간의 합으로 계산

4) 기울기 ((C−B) 대 (B−1))



$$P_3 = 0.60\,P_2 \pm 0.01\,P_2$$